IMPERIAL COLLEGE LONDON

DEPARTMENT OF ELECTRICAL AND ELECTRONIC ENGINEERING
EXAMINATIONS

EEE PART IV: MEng and ACGI

HVDC TECHNOLOGY AND CONTROL

Tuesday, 12 May 2026, 10:00 AM

Time allowed: 3:00 hours

There are THREE questions on this paper.

Answer ALL questions.

Questions 1 & 2 carry 25 marks each, Question 3 carries 30 marks.

Any special instructions for invigilators and information for candidates are on page 1.

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Question 1

a) The inverter end of an LCC-HVDC inverter is connected to an AC system whose strength can be characterised using a Thevenin equivalent reactance X_{eq} at the point of connection. Using an appropriate network diagram, discuss why and how the operation of the HVDC link depends on the strength of the AC system.

[5]

b) For a point-to-point LCC-HVDC link, different control objectives are assigned to the two converter stations at the two ends. Why under normal operation, the rectifier station controls the direct current, while the inverter station controls the direct voltage and when that role can reverse?

[4]

c) Using qualitative reasoning only, explain how a reduction in AC-side voltage at one end can indirectly increase reactive power demand at the opposite end of an LCC-HVDC link. Your answer should separately consider reduction of voltage at:

i) Rectifier-side AC system

[3]

ii) Inverter-side AC system

[3]

d) Two separate 275 kV AC systems are to be interconnected in two stages. The short-circuit level at respective the points of connection are 15,000 MVA and 12,000 MVA, respectively. In first stage of interconnection a 275 kV AC transmission line with 5Ω reactance is commissioned. In the second stage, a bipole point-to-point LCC-HVDC link is to be built. Each converter station of the LCC-HVDC links is designed to absorb reactive power equal to 65% of the active power transmitted through the DC link. The minimum acceptable effective short-circuit ratio (ESCR) is 3. The AC systems and the AC line may be modelled as purely reactance neglecting any resistance.

i) Determine the maximum permissible rated active power capacity (in GW) of the LCC-HVDC link that satisfies the ESCR requirement at both ends.

[7]

ii) How much active power can be transmitted through the LCC-HVDC link without violating the ESCR requirement when the AC line is out of operation?

[3]

Question 2

Consider a point-to-point bipole LCC-HVDC link whose rating and parameters are given in Table 2.1. The link interconnects two different 400 kV AC systems. Under normal operation, the rectifier station regulates the direct current at its rated value, while the inverter station maintains the rated direct voltage.

Rated direct voltage	$\pm$500 kV
Rated power	2000 MW
Nominal gate delay angle	17°
Minimum gate delay angle	4°
Minimum extinction advance angle	20°
DC resistance per pole	4 Ω
Commutation resistance per 6-pulse converter	4 Ω
Current margin	15%

Table 2.1: LCC-HVDC link parameters

Using *per-pole calculations*, determine the effective transformer turns ratios and the reactive power (MVAr) absorbed at both rectifier end and at the inverter end under each of the following operating conditions:

a) Both AC systems operating at their rated voltages.

[10]

b) The AC voltage at the inverter terminal reduced to 92% of its rated value, while the rectifier-side AC voltage remains at its rated value.

[5]

c) The AC voltage at the rectifier terminal reduced to 88% of its rated value, with the inverter-side AC voltage at its rated value. If the rectifier reaches its minimum firing delay angle, assume it continues operating at this limit.

[10]

Question 3

Consider a basic one-arm modular converter connected between an AC grid and a DC grid represented as equivalent voltage sources as shown in Figure 3.1. The parameters of the converter are given in Table 3.1. Answer the questions below.

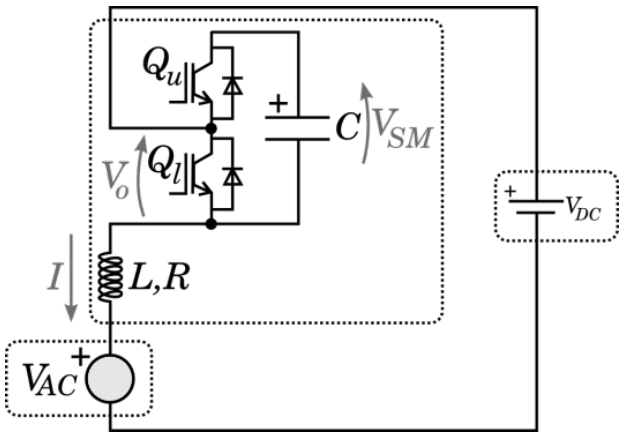


Figure 3.1 – Diagram of a one-arm modular AC/DC converter

Parameter	Value	Units	Brief description
V_{DC}	700	V	Rated DC grid voltage.
V_{AC}	400	V rms	Rated AC grid voltage
V_{SM}	1.4	kV	Submodule capacitor voltage
S_{rated}	200	kVA	Rated power of the converter
$f_{nominal}$	50	Hz	Nominal frequency of the AC grid
R	24	mΩ	Equivalent resistance of bridge plus the filter
X (from L)	160	mΩ	Reactance of the filter at nominal grid frequency.

Table 3.1 – Parameters of the one-arm modular AC/DC converter

a) Consider the converter working at zero current and answer the following questions
   i) What are the AC and DC components of the output voltage of the arm, V_o,
   [2]
   ii) What is the minimum submodule capacitor voltage, V_{SM}, required to generate this voltage?
   [2]
   iii) How low could the DC grid voltage be before the converter became unable to control the current?
   [2]

b) List the advantages and the disadvantages or replacing the half-bridge submodule in this design by a full-bridge submodule with the same submodule capacitor voltage.
   [3]

c) Consider the converter operating with zero net power balance across the submodule (ie its energy doesn't drift over time) and 200 kW of active power being delivered to the AC grid with unity power factor (P_{AC}=200 kW, Q_{AC}=0). Answer the following questions and highlight any assumptions or simplifications you make.
   i) Explain what condition must be met for the submodule to have zero net power balance (ie what the implications are for the AC and the DC voltages and currents of the converter arm).
   [2]
   ii) What are the DC and the AC components of the output voltage of the submodule and the current required?

[5]

d) Consider an operating point where I_{AC}=100 A (taking V_{AC} as the reference of angles) and I_{DC}=200 A (flowing towards the submodule from the DC grid) and answer the following questions:

i) What is the active and reactive power of the AC grid and the power of the DC grid?

[1]

ii) What are the DC and AC components of the output voltage of the stack?

[2]

iii) What is the active and reactive power of the submodule and what is the net power of the submodule?

[3]

e) Consider a current controller design where we use a PI with

$$K_P = \frac{-L}{\tau^*} = -0.17 \text{ V/A} \qquad \text{and} \qquad K_P = \frac{-R}{\tau^*} = -8 \text{ V/(A s)}$$

Note that we chose $\tau^* = 0.003$ s to get a closed-loop response equivalent to a 1st order system with a time constant of 3 ms. Complete the following tasks:

i) Write the open-loop transfer function between V_o and I and explain how it is obtained.

[1]

ii) Write the closed-loop transfer function between I^* (current reference) and I and explain how it is obtained.

[2]

iii) What would be the phase delay (compared to the reference) and the amplitude of the AC current if we gave the current controller a reference of 1A of peak amplitude at 50 Hz?

[3]

iv) Write the transfer function between the AC grid-voltage and the current (assume the closed-loop current controller is used and there's no feed-forward controller)

[2]